European Polymer Journal 222 (2025) 113594

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Biobased and biodegradable polyester amides based on nylon 6,6 and polybutylene adipate via straightforward bulk polymerization

Cornelis Post [a,b], Jeroen van der Vlist [c], Jesse A. Jongstra [d], Rudy Folkersma [b], Vincent S.D. Voet [b], Katja Loos [a,*]

[a] Macromolecular Chemistry & New Polymeric Materials, University of Groningen, Nijenborgh 3, 9747 AG Groningen, the Netherlands
[b] Circular Plastics, NHL Stenden University of Applied Sciences, Van Schaikweg 94, 7811 KL Emmen, the Netherlands
[c] Senbis Polymer Innovations B.V., Eerste Bokslootweg 17, 7821 AT Emmen, the Netherlands
[d] Department of Chemical Engineering, Engineering and Technology Institute Groningen (ENTEG), University of Groningen, Nijenborgh 3, 9747 AG Groningen, the Netherlands

ARTICLE INFO

Keywords:
Polyester amide
Biobased
Biodegradation
Nylon 6,6
Bulk polymerization

ABSTRACT

Aliphatic polyesters are generally known for their limited thermal and mechanical properties, but are prone to biodegradation. On the other hand, aliphatic polyamides, mainly represented by nylons, feature high thermal transition points and enhanced mechanical strength, but are usually classified as nonbiodegradable. In this work, the properties of the aliphatic polyester polybutylene adipate and nylon 6,6 were combined by synthesizing polyester amides. Relatively low- and high-molecular weight series, $\overline{M_n}$ 7000 and 14400 g/mol, respectively, were synthesized via a straightforward bulk polymerization process by utilizing biobased starting materials: butanediol, adipic acid and nylon 6,6 salt. The amide-to-ester ratio was varied to analyze the influence of the amide content on the molecular structure, thermal properties, mechanical properties, hydrophilicity and biodegradability. Strong relationships between the amide content and the T_g, T_m, degree of crystallinity, and mechanical properties were observed. Water contact angle measurements revealed that all the polyester amides were hydrophilic. Finally, the biodegradation experiments demonstrated that all polyester amides are prone to biodegradation in activated sludge and that the amide content had a major influence on the biodegradation rate. This work supports the high potential of biobased polyester amides for applications in numerous fields due to their versatility in terms of polymeric properties and sustainable character.

1. Introduction

In recent decades, biobased polymers have drawn attention from both academia and industry to find more sustainable alternatives to replace conventional dominant fossil-based polymers[1,2]. Biobased polymers are derived from renewable resources such as agricultural products and organic waste streams[3]. Generally, biobased polymers, such as polylactic acid (PLA) and bioethylene-based polymers, have a lower carbon footprint, in terms of greenhouse gas emission, compared to their fossil based alternatives[4]. However, greenhouse gas emissions are not the only current threat to fossil-based commodity plastics. Polyethylene, polypropylene, polystyrene, polyvinyl chloride, polyurethanes and polyethylene terephthalate take account for 75 % of the plastic production worldwide and are resistant to biodegradation [5–7]. Improper waste management or dumping of these plastics in landfills can therefore lead to marine, fresh water and soil pollution [8–10]. However, their accumulation in the environment is not the main pitfall of their resistance to biodegradation since adequate waste handling and recycling of these materials could limit this. A growing concern of nonbiodegradable polymers is the formation of microplastics. These environmental contaminants are smaller than 5 mm and originates from different sources, including personal care products, paints, washing wastewater, rubber tires, and macroplastics (>5 mm), such as bottles and bags, which can break into smaller pieces[11,12]. These microplastic have properties similar to those of their original products, i.e., high stability and long residence time, and might end up in the food chain and lead to exposure to toxic contaminants and additives[13]. This means that even during use and proper waste management, plastic particles might still end up in the environment and could lead to serious threats. Therefore, an optimum should be found between polymer

* Corresponding author.
E-mail address: k.u.loos@rug.nl (K. Loos).

https://doi.org/10.1016/j.eurpolymj.2024.113594
Received 5 September 2024; Received in revised form 18 November 2024; Accepted 21 November 2024
Available online 22 November 2024
0014-3057/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

performance, durability and the rate of biodegradation.

Aliphatic polyesters are a class of polymers that can be enzymatically hydrolyzed and some polymers are prone to biodegradation[14,15]. Examples include polybutylene succinate (PBS) and polycaprolactone (PCL), which are used for agricultural and biomedical applications [16–18]. However, aliphatic polyesters are prone to hydrolytic degradation, have generally limited thermal and mechanical properties, require additives or need to be blended with other polymers, although it is dependent on the application [16,19].

On the other hand, polyamides (PAs), such as nylon 6 and nylon 6,6, are known for their high mechanical strength, chemical resistance and thermal properties[20,21]. Although synthetic polyamides are classified as nonbiodegradable, polyamides are present in nature, for example, in wool and silk and in proteins, which are biodegradable[21,22].

Polyester amides (PEAs) have both ester and amide functional groups in the polymer chain and therefore show good thermal and mechanical properties due to the hydrogen bonding of the amide groups [23]. In addition, the properties of polyester amides can be tuned by varying the type of monomer and the amide-to-ester ratio[24,25]. Therefore, polyester amides can combine the biodegradability of aliphatic polyesters and the enhanced thermal and mechanical properties of polyamides[25,26].

Liu *et al.* (2011) synthesized a "prePA" from caprolactam and 2-aminoethanol, which subsequently reacted with adipic acid and 1,4-butanediol via melt polycondensation at different ratios to vary the amide content[27]. Similar work was performed by Hao *et al.* (2013), who made a "prePA" from nylon-6,10 salt and sebacic acid, which was also polymerized by using adipic acid and butanediol in different ratios to vary the polyamide segments[28]. In addition, Armelin *et al.* (2002) synthesized polyester amides, based on nylon 6,10 or nylon 12,10 units, via interfacial polymerization using sebacoyl dichloride, an aliphatic diol and a diamine, respectively[29]. The German company Bayer was also interested in polyester amides based on nylon repeating units and patented this in 1997[30,31].

In this work, polyester amides were synthesized from adipic acid, 1,4-butanediol and nylon 6,6 salt, which is also known as AH salt. All these monomers can be biobased, and several companies (Covestro, DSM and Toray) are working at the commercialization of these starting material from renewable resources at relatively large scale [32–34]. The advantages of the use of salt with respect to the individual constituents, adipic acid and hexamethylenediamine, are the equal stoichiometry and avoidance of the use of a highly volatile and low melting point monomer (hexamethylenediamine, 42 °C)[35]. Furthermore, the polyester amides in this work were synthesized in a single step via a straightforward bulk polycondensation reaction, which is preferred by industry and is more sustainable than the use of highly reactive diacyl chlorides[29]. Since the trade-off between polymer performance and biodegradation varies depending on the application, synthesizing a broad variety of distinct polyester amides offers an appealing overview of the wide range of potential utilization. Hence, the amide-to-ester ratio, i.e., amide content, was varied by using different feed ratios between the polyester monomers (adipic acid and butanediol) and polyamide monomers (nylon 6,6 salt). These polymers were analyzed by ^{1}H NMR, GPC, TGA, DSC, WAXD, tensile tests, water contact angle measurements and tested for their biodegradability. This demonstrates a detailed overview of the structural, thermal, physical and mechanical properties of a broad range of polyester amides.

2. Experimental

2.1. Materials

Hexamethylenediamine (HMDA, 98 %), adipic acid (AA, 99 %), butanediol (BD, 99 %), titanium isopropoxide (99.99 %), and dimethylformamide (DMF, HPLC grade 99,99 %) were all purchased from Sigma-Aldrich. Chloroform (HPLC grade) was obtained from Macron Fine Chemicals and diethyl ether (technical grade) was supplied by Honeywell Research Chemicals. All chemicals were used as received, without any treatment or purification. The wastewater treatment facility in Glimmen, the Netherlands, kindly supplied the activated sludge.

2.2. Synthesis of AH salt

The adipic acid hexamethylene diamine salt (AH salt) procedure was derived from a method described by Beck and Magat[36]. First, 29.20 g (0.200 mol) of adipic acid was dissolved in 250 mL of stirring ethanol and slowly heated until it dissolved. When the AA solution was allowed to slowly cool, 23.47 g (0.202 mol) of HMDA was dissolved in 50 mL of ethanol. The HMDA ethanol solution was slowly poured into the AA ethanol solution, followed by precipitation of the AH salt. The mixture was centrifuged (4500 RPM, 10 min) and the ethanol was poured off. The product was allowed to dry in a fume hood for several days, analyzed by TGA and DSC, and obtained as a white solid in 98 % yield.

2.3. Synthesis of polyester amides

The melt polycondensation reactions were performed in a 25 mL round bottom flask equipped with a magnetic stirrer and connected to a distillation setup. The system was placed either under argon flow or high vacuum via a Schlenk line. The polymerizations were conducted at a 10

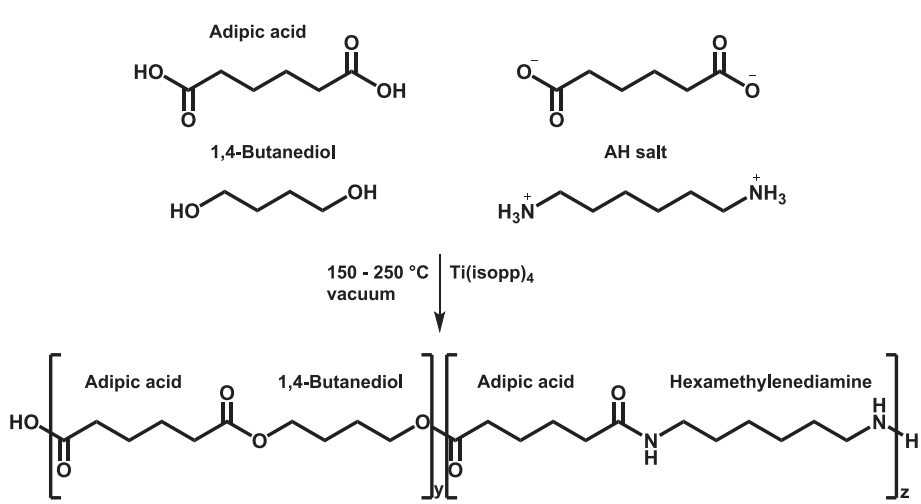

Fig. 1. Schematic overview of the synthesis of a polyester amide from adipic acid, butanediol and the AH salt, wherein the ratio of the ester(Y) and amide(Z) segments can be varied.

g scale, and the monomer molar ratios were varied to synthesize polyester amides with different amide contents. The catalyst titanium isopropoxide (600 ppm), butanediol (1.01 eq.), adipic acid (1.00 eq.) and AH salt (0.0 to 1.0 eq.) were added directly to the flask in the order indicated. The setup was placed under a mild argon flow and heated to 150 °C for 1 h. Water evolved as condensate and was removed via the distillation setup. Subsequently, the temperature was increased to 200 °C and maintained for 2 h. Then, the system was gently switched to vacuum ($2_{*}10^{-2}$ mbar) and kept at 200 °C for another 2 h. Finally, the temperature was increased to 250 °C and maintained for 3 h. The stirring speed was reduced from 300 rpm to a minimum of 40 rpm during the process to maintain regular stirring behavior while the viscosity of the medium increased.

A similar process was used for a second series of polyester amides with equal amide-to-ester ratios, but with an extension of reaction time with the aim to increase the molecular weight. In this second series, the last step (250 °C, vacuum) was maintained for 2 h, stopped during the night, and continued for another 7 h the next day.

The obtained polyester amides were dissolved/gelated in chloroform and precipitated in diethyl ether. The products were isolated via centrifugation (4500 rpm, 10 min) and dried in a fume hood. The polyester amides were designated as PEA_X_Y, based on the amide content derived from ^{1}H NMR (X = 0 – 50 %) and series (Y = 1 or 2). For example, PEA_30_2 is part of the second series and has an amide content of 30 %. An overview of the polymerization route is depicted in Fig. 1.

2.4. Poly(1,4-butylene adipate-co-[imino(1,6-dioxohexamethylene) iminohexamethylene])

The yields of the polyester amides range from 88 to 96 % and are based on gravimetric analysis. ^{1}H NMR (400 MHz, DMSO-d_6), δ (ppm): 7.71 (s, 2H, N**H**), 3.99 (m, 4H, ester-**CH_2**, BD), 2.97 (m, 4H, amide-**CH_2**, HMDA), 2,27 (m, 6H, ester-**CH_2**, AA), 2.02 (m, 2H, –**CH_2**-amide, AA), 1.58–1.20 (m, remaining aliphatic protons).

2.5. Characterization

A 400 MHz Varian VXR spectrometer was used for proton nuclear magnetic resonance (^{1}H NMR) measurements using DMSO-D_6 as the solvent. The software MestReNova was used to process the obtained spectra.

Gel permeation chromatography (GPC) was carried out to determine the molecular weights of the synthesized polyester amides. A Viscotek GPCmax, equipped with model 302 TDA detectors and two PolarGel L and M columns (8 μm × 300 mm) from Agilent Technologies, was used to perform the analyses. The system was operated at 50 °C, and DMF, containing 0.01 M LiBr, was used as the solvent and eluent at a flow rate of 1.0 mL/min. A conventional calibration curve was derived from narrow dispersity poly(methyl methacrylate) (PMMA) standards (Polymer Standards Service, $\overline{M_w}$ ranging from 640 to 1 100 000 g/mol) and used to determine the number average molecular weight and weight average molecular weight. The software Viscotek OmniSec (version 5.0.) was used to process the data. The samples were prepared at a concentration of 2 mg/ml and filtered through a 0.20 μm PTFE filter prior to injection.

Not all the obtained polyester amides were soluble in DMF, so in addition all the polyester amides were subjected to GPC using 1,1,1,3,3,3-hexafluoroisopropanol (HFIP) with 0.019 % NaTFA salt as the solvent and eluent. The GPC apparatus consisted of two PFG combination medium microcolumns (7 μm particle size, 4.6 mm × 250 mm, separation range 100 − 1.000.000 Da), a precolumn PFG combination medium (7 μm particle size, 4.6 mm × 30 mm) and a refractive index detector (RI). The polyester amide samples were prepared by dissolving 5.0 mg of the polymer in 1.5 mL of the solvent and were filtered through a 0.2 μm PTFE syringe filter prior to injection.

Thermogravimetric analysis (TGA) was conducted to estimate the thermal stability of the polyester amides on a TA-Instruments Discovery TGA 5500 device. The sample sizes were in the order of a few mg and the samples were heated at a constant rate of 10 °C/min to 700 °C under an inert atmosphere (N_2).

The glass transition temperatures (T_g) and melting points (T_m) of the polyester amides were determined via differential scanning calorimetry (DSC) analysis. A TA-Instruments Q1000 DSC was used with a heating and cooling rate of 10 °C/min under a mild nitrogen flow, and a sample size of approximately 5 mg was used. The temperature ranged from −70 to 250 °C for two heating ramps and one cooling ramp.

Wide-angle X-ray diffraction (WAXD) analysis was used to determine the relative degree of crystallinity of the processed polyester amides. The top and bottom parts of the tensile test specimens were used for this analysis; more details are provided below. A Bruker D8 Endeavor diffractometer ($Cu_{K\alpha}$ radiation, λ = 0.1541 nm, 40 kV, 25 mA) was used and operated in the angular range (2θ) from 4 to 51° at ambient conditions. The obtained WAXD spectra were processed with the software Origin 2018 to separate the crystalline and amorphous peaks via peak deconvolution. Subsequently, the degree of crystallinity (χ_c) was calculated from Equation (1), where $Area_c$ and $Area_a$ are the cumulative integration values of the crystalline and amorphous peaks, respectively.

$$\chi_c(\%) = \frac{Area_C}{Area_A + Area_C} x100 \quad (1)$$

The mechanical properties of the polyester amide series were analyzed via an Instron tensile tester 4301, equipped with a 1000 N load cell. The specimens were prepared via compression molding using a Fontijne LabManual 300 hot press at 30 °C above the melting point of the polymer for 10 min with a force of 50 kN. The samples were allowed to cool slowly in the press without pressure applied. The mold for the dumbbell-shaped tensile bars had the following size: 38 mm overall length, 16 mm overall width, 15 mm gauge length, 4.6 mm width of narrow section, and a thickness of 2.0 mm. The width of the narrow section and thickness were measured prior to every measurement due to possible shrinkage or deformation of the samples over time. Each unique polyester amide was measured in quintuplicate at a strain rate of 1 mm/min at room temperature. The data were averaged and used to determine the ultimate stress (σ_m), strain at ultimate stress (ε_m) and Young's modulus (E).

The top and bottom parts of the used tensile test specimens were reprocessed into flat films of approximately 1 mm in thickness via hot pressing, at similar conditions as mentioned above. These flat polymers films were used for water contact angle measurements on a Dataphysics OCA 15EC device. The software SCA20 was used to take pictures and to determine the left, right and mean water contact angles. The water droplet (5.00 μL, dispensing rate of 5.00 μL/s) was photographed after several seconds when the droplet and surface equilibrated. Each polymer sample was measured in quintuplicate and averaged.

The biodegradability of the polyester amides was analyzed by using activated sludge and a Lovibond® Water Testing BOD-System BD 600 setup. The experiments were conducted according to the OECD 301F manometric respirometry test guidelines[37]. The procedure was the same as that in our previous work[38]. Briefly, biodegradation analysis is based on the biological oxygen demand (BOD), which ranges from 0 to 400 mg/L oxygen. The theoretical oxygen demand (ThOD) of each polyester amide was calculated and used to derive the sample mass required to reach a maximum BOD of 400 mg/L oxygen. The polyester amide products were sieved, and only the fraction with a particle size between 500 and 800 μm was used for the biodegradation analysis to limit differences in particle size and surface area. The test medium consisted of diluted activated sludge containing various nutrient salts. Each polymer sample, a control and reference (sodium acetate), was measured in duplicate for 28 days in a dark environment at room temperature. The degree of biodegradability is expressed as the percentage

of the maximum BOD value relative to the BOD value of the control:

$$\text{Biodegradation}\,(\%) = \frac{BOD_{testsample} - BOD_{control}}{400} * 100 \quad (2)$$

3. Results and Discussion

3.1. Synthesis and characterization

Two series of polyester amides with an increasing amide-to-ester ratios and amide contents were successfully synthesized via a straightforward one-pot bulk polymerization process. The commercially available catalyst titanium isopropoxide, which is widely used in polyester syntheses and for the production of PEAs, was used[39,40]. This process is advantageous compared with other synthesis methods for the production of similar PEAs, such as interfacial polymerization and multistep synthesis via either prePAs or prePEAs, due to its simplicity, scalability and efficiency[27–29]. In addition, the solvent free condition improves the sustainability of the process. Bulk polymerization at elevated temperature and under high vacuum was possible due to the use of AH salt, which would not be feasible when the volatile diamine was used directly [35]. The success of this process was demonstrated by the high yields, ranging from 88 to 96 %.

The structural composition of the synthesized polyester amides was confirmed by using ^{1}H NMR analysis. The PEA_35_2 polymer is taken as an example, and its spectrum is shown in Fig. 2. All signals could be assigned to the expected monomer units in the polymer structure, similar to those of polyester amides based on butanediol, adipic acid and ε-caprolactam[40]. The actual amide-to-ester ratios and amide content of the polymers were determined by analyzing the distinctly visible protons of the butanediol residues ("5,8") and the hexamethylene diamine segments ("14,19"). The results are displayed in Table 1 and Table 2, and Figure S1 and Figure S2 in the Supporting Information, for the low and high molecular weight series, respectively. In both series, the NH_2: OH feed ratios were very similar to the actual amide-to-ester ratios observed in the polyester amides. This illustrates the absence of any composition drift at high conversion and full utilization of the monomers, which is an advantage compared to other polyester amide synthesis routes where sometimes an excess of one of the starting materials is required[29,41].

The molecular structures of the polyester amides were further characterized by GPC. The number average molecular weights ($\overline{M_n}$) and weight average molecular weights ($\overline{M_w}$) of all synthesized polyester amides, measured in DMF and/or HFIP, are given in Table 1 and Table 2, and illustrated in the Supporting Information. The polyester amides with an amide-to-ester ratio up to 0.5: 1.0 (amide content of 35 %) were soluble in DMF, while the complete series were soluble in HFIP and no other common organic solvents. This illustrates the general resistance of polyester amides to chemical solvents, which is beneficial for numerous applications. The average $\overline{M_n}$ values of the first series and second series were 7000 g/mol and 14 400 g/mol, respectively, and are therefore referred to as the low and high molecular weight series, respectively. These series demonstrated that the chain length can be increased by prolonging the reaction time, typical for step-growth polymerizations. However, the molecular weights remained relatively low, likely due to the inefficient mixing behavior of the highly viscous polymer melt at high conversion levels. This issue restricts the progress of the reaction, but it could be mitigated by replacing the magnetic stirrer with a mechanical stirring system.

In both separate series, the synthesis procedures were identical, which resulted in similar molecular weights for the polyester amides and were independent of the amide content. Only the pure polyester (polybutylene adipate) (PBA) slightly outperformed the other polymers in the high $\overline{M_n}$ series. The molecular weights from the DMF GPC analysis were slightly higher than those from the HFIP GPC analysis, and these differences were attributed to variations in the polymer–solvent interactions, columns and calibration curves. For example, the LiBr in DMF or the polarity of HFIP are expected to influence the hydrodynamic radius and volume and therefore the behavior and interactions with the column material. The data of the polymers from the high $\overline{M_n}$ series will be further elaborated in the next paragraphs, and the results of the low $\overline{M_n}$ series can be found in the Supporting Information.

3.2. Thermal behavior and crystallization

The thermal properties of the polyester amides were analyzed using TGA and DSC analysis. A polyester amide can have a random, alternating, or segmented (block) structure. Polyester amides with block structures usually have better thermal properties, higher thermal stability and enhanced mechanical properties due to the crystalline amide-rich hard phase, which acts as a thermoreversible physical crosslinker with the amorphous phase[28,42]. However, a homogeneous random polyester amide in the absence of rigid nylon segments, which are not prone to biodegradation, is preferred in this study. The thermal stabilities of the synthesized polymers are represented by the $T_{d5\%}$ values and are given in Table 1 and Table 2 for the low and high $\overline{M_n}$ series, respectively. All polyester amides have a high thermal stability, are in a

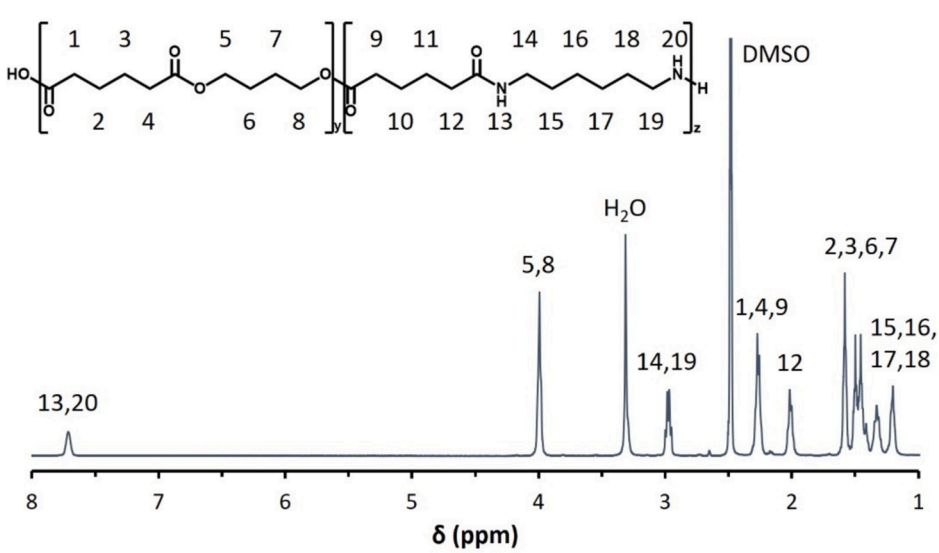

Fig. 2. ^{1}H NMR spectrum of a polyester amide (PEA_35_2) with an amide content of 35 % (amide-to-ester ratio of 0.5: 1.0) in DMSO-d_6.

Table 1
Structural and thermal characteristics of the low molecular weight polyester amides.

Polymer	NH_2:OH (Feed)	Amide/Ester (^{1}H NMR)	Amide (%)	$T_{d5\%}$ (°C)	T_g (°C)	$T_{m,1}$ (°C)	$T_{m,2}$ (°C)	$T_{m,3}$ (°C)	$\overline{M_n}$ DMF (g/mol)	$\overline{M_w}$ DMF (g/mol)	$\overline{M_n}$ HFIP (g/mol)	$\overline{M_w}$ HFIP (g/mol)
PEA_00_1	0.0: 1.0	0.00	0	338	−55	57	–	–	7500	19,200	10,000	19,600
PEA_10_1	0.1: 1.0	0.11	10	353	−53	54	95	–	8800	23,700	9700	20,200
PEA_18_1	0.2: 1.0	0.22	18	348	−49	64	98	128	8100	23,100	8500	19,000
PEA_25_1	0.3: 1.0	0.34	25	345	−47	67	106	133	7300	16,400	6000	14,500
PEA_30_1	0.4: 1.0	0.44	30	350	−42	66	128	149	6800	14,900	5700	13,300
PEA_35_1	0.5: 1.0	0.54	35	352	−38	66	138	172	7800	17,500	6700	15,200
PEA_37_1	0.6: 1.0	0.58	37	359	−34	70	143	178	–	–	5400	13,500
PEA_40_1	0.7: 1.0	0.67	40	357	−30	–	150	184	–	–	6800	16,800
PEA_44_1	0.8: 1.0	0.78	44	357	−28	–	154	191	–	–	6000	14,000
PEA_47_1	0.9: 1.0	0.90	47	363	−24	–	156	197	–	–	6500	15,500
PEA_50_1	1.0: 1.0	1.00	50	360	−21	–	154	201	–	–	5800	13,900

Table 2
Structural and thermal characteristics of the high molecular weight polyester amides.

Polymer	NH_2:OH (Feed)	Amide/Ester (^{1}H NMR)	Amide (%)	$T_{d5\%}$ (°C)	T_g (°C)	$T_{m,1}$ (°C)	$T_{m,2}$ (°C)	$T_{m,3}$ (°C)	$\overline{M_n}$ DMF (g/mol)	$\overline{M_w}$ DMF (g/mol)	$\overline{M_n}$ HFIP (g/mol)	$\overline{M_w}$ HFIP (g/mol)
PEA_00_2	0.0: 1.0	0.00	0	326	−58	56	–	–	25,400	47,400	23,900	47,500
PEA_09_2	0.1: 1.0	0.10	9	371	−51	55	66	95	16,400	31,000	13,900	28,200
PEA_18_2	0.2: 1.0	0.22	18	361	−51	63	108	125	16,300	34,500	14,700	31,700
PEA_23_2	0.3: 1.0	0.31	23	373	−50	66	109	140	17,700	35,700	15,200	31,000
PEA_30_2	0.4: 1.0	0.43	30	362	−43	67	124	147	19,800	39,300	14,900	37,600
PEA_35_2	0.5: 1.0	0.54	35	361	−40	72	137	169	15,500	33,400	14,300	39,000
PEA_37_2	0.6: 1.0	0.59	37	363	−34	–	142	177	–	–	13,100	30,400
PEA_41_2	0.7: 1.0	0.70	41	366	–32	–	147	181	–	–	10,700	25,500
PEA_44_2	0.8: 1.0	0.78	44	355	−29	–	150	188	–	–	11,300	26,700
PEA_48_2	0.9: 1.0	0.92	48	364	–22	–	151	197	–	–	14,000	29,800
PEA_50_2	1.0: 1.0	0.98	50	359	−19	–	155	201	–	–	12,300	30,000

similar range (345 – 373 °C) and not influenced by the amide content. Although the pure polyesters had slightly lower values, they still demonstrated high thermal stability. The observed thermal stabilities of these polyester amides were in good agreement with those of similar polyester amides reported previously[27,29].

The glass transition temperatures, measured by DSC, of the high $\overline{M_n}$ polyester amide series are illustrated in Fig. 3. For all synthesized polyester amides in this series, only a single T_g transition point was observed. This indicates that the amorphous phases of the polyester amides consisted of a single homogeneous phase, which is in accordance with the literature on similar polyester amides[27,41]. The T_g of the pure polyester (PBA) was observed at −58 °C, which is similar to the reported value (−52 °C) by Lips *et al.* (2005), based on dynamic mechanical analysis (DMA)[42]. Noteworthy is the dependence of the T_g on the amide content of the polyester amides (Fig. 3). The T_g is at a minimum for the lowest amide-to-ester ratio of 0.1: 1.0 (amide content of 9 %) and then increases linearly. The gradual linear increase, from −51 to −19 °C, was expected due to the increased rigid amide content resulting from hydrogen bonding[28]. Nevertheless, the observed T_g values are still well below the T_g of pure polyamide, Nylon 6,6, which is around 53 °C (DSC)[43].

The low T_g values of the polyester amides might indicate less favorable interactions between the polymer chains, i.e., limited hydrogen bonding possible yet in the amorphous phase, which leads to a larger extent of chain mobility and a higher degree of freedom.

The melting endotherms of the pure polyester and the polyester amides are depicted in Fig. 4. The pure polyester PBA has a main melting point of 56 °C, which corresponds well with a previously reported study (55.6 °C)[44]. All synthesized polyester amides demonstrated multiple broad and overlapping melting endotherms, indicating different crystalline phases. Up to an amide content of 37 %, the crystalline phases of the pure polyester segments were still present, as revealed by the appearance of the melting points around 55 to 70 °C. The magnitude of these polyester crystalline phases decreased with increasing amide content (Figure S7 Supporting Information) and were practically not visible anymore for the polyester amides with an amide-to-ester ratio of

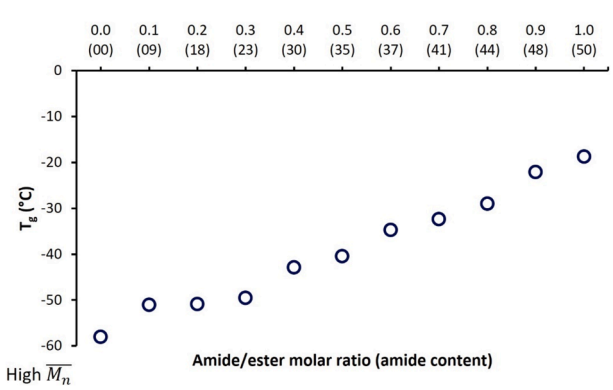

Fig. 3. Glass transition temperature (T_g) versus the amide-to-ester ratio and amide content (%) of the high $\overline{M_n}$ polyester amide series.

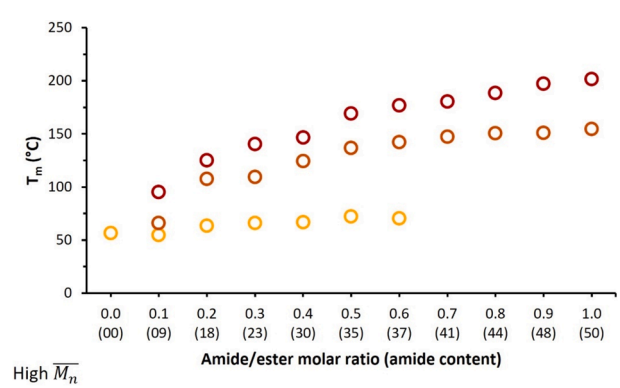

Fig. 4. Melting endotherms (T_m) versus the amide-to-ester ratio and amide content (%) of the high $\overline{M_n}$ polyester amide series.

0.7: 1.0 (amide content of 41 %) and higher.

In addition, the melting points of the polyester amides, ranging from 66 to 201 °C, increased with increasing amide content, as expected, but did not reach the melting point of Nylon 6,6 (263 °C)[43]. This indicated that pure polyamide segments were absent in the polyester amides. Noteworthy, generally all polyester amides demonstrated two different broad melting peaks, both increasing for higher amide contents. This indicates the presence of two different and varying crystalline ester-amide phases. This phenomenon is also observed in similar aliphatic polyester amides and has been linked to the average ester-amide sequence length, which leads to variations in crystal structure[29,42].

Finally, an additional note has to be made about PEA_18_2, which showed besides the regular polyester and polyester amide peaks, an extra crystallization exotherm (−17 °C) and melting endotherm (20 °C). These peaks are also observed for the same polyester amide from the low $\overline{M_n}$ series, but not for any other sample. This phenomenon, which could be attributed to an additional polyester amide crystalline segment consisting of weak interactions, was also observed by Lips et al. (2005) for a polyester amide consisting of 10 % amide bonds[42].

The DSC curves, represented in the Supporting Information Figure S7, indicated differences in the degree of crystallinity between the polyester amides, based on the intensities of the melting endotherms. WAXD analysis was performed to support this hypothesis, and the normalized WAXD patterns of the high $\overline{M_n}$ series are illustrated in Fig. 5. The appearance of the clear crystalline peaks, at 2θ values of 21.7 and 22.4°, for the pure polyester PBA is a major observation. Those peaks were still visible for the polyester amide containing the lowest amide content (PEA_09_2), but considerably different patterns were observed for the other polyester amides. These polyester amides have two major crystalline peaks at 2θ values of approximately 20 and 24°. This is in correspondence with similar polyester amides based on butanediol, dimethyl adipate and aliphatic amide units, and these signals correspond to the diffraction pattern of a crystal lattice caused by intermolecular amide-amide hydrogen bonding[28,42].

Furthermore, the intensities of these two crystalline peaks increased with increasing amide content. This phenomenon is better visualized in Fig. 6, where the degree of crystallinity of both the low and high $\overline{M_n}$ series are plotted versus the amide-to-ester ratio and amide content. The crystalline peaks were separated from the amorphous parts via peak deconvolution to determine the degree of crystallinity; as depicted in the Supporting Information. Fig. 6 illustrates the significantly higher degree of crystallinity of the pure polyesters (52–56 %) in both the low and high $\overline{M_n}$ series. Subsequently, the degree of crystallinity decreased and reached a minimum (11 %) for the polyester amide containing an amide content of 18 %, and then gradually increased (up to 23 %) with increasing amide content. This observation is in line with expectations that a higher amide content results in a more efficient packing of the chains, i.e., a higher degree of crystallinity.

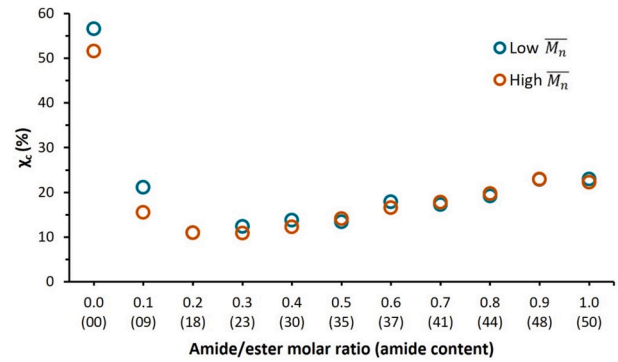

Fig. 6. Degree of crystallinity versus the amide-to-ester ratio and amide content (%) of both the low and high $\overline{M_n}$ of the polyester amide series, obtained from the WAXD patterns.

The crystallization behavior of the polyester amides is similar to the relationship between the T_g and the amide content, as represented in Fig. 3. This implies that the amide content in the polyester amides has a strong influence on the intermolecular interactions caused by hydrogen bonding between the amide groups on the chain mobility, flexibility and crystallization behavior.

4. Mechanical Analysis

The influence of the amide content of the polyester amides on the mechanical properties was analyzed by tensile testing, and the results are shown in Table 3 and some representative stress–strain curves are illustrated in the Supporting Information. The ultimate stress (σ_m), the highest stress value reached during the tensile test, ranged from 6.3 MPa to 15.9 MPa. Surprisingly, the highest stress values were observed for the pure polyester. This was attributed to the significantly higher degree of crystallinity and to a lesser extent the slightly increased molecular weight of PBA compared to all other polyester amides. Polymers containing a higher degree of crystallinity generally exhibit a higher tensile strength, enhanced Young's modulus and lower strain at break due to the increased rigidity and lower mobility of the chains[45]. Besides the pure polyester, an optimum ultimate stress was observed for the polyester amides having an amide-to-ester ratio of 0.5: 1.0 (amide content of

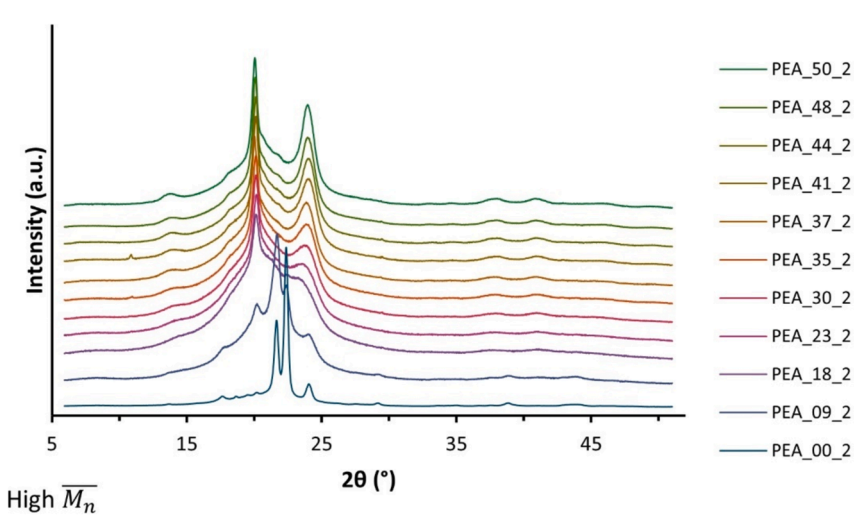

Fig. 5. Normalized WAXD patterns of the high $\overline{M_n}$ polyester amide series.

Table 3
Ultimate stress, strain at ultimate stress and E-moduli of the high $\overline{M_n}$ polyester amide series.

Polymer	Ultimate Stress (MPa)	STD (MPa)	Strain (%)	STD (%)	E-Modulus (MPa)	STD (MPa)
PEA_00_2	15.9	1.7	10.4	3.2	267.4	6.0
PEA_09_2	6.8	0.5	6.9	0.5	166.3	4.9
PEA_18_2	8.4	0.4	233.2	32.3	87.7	2.5
PEA_23_2	9.6	0.4	23.2	3.3	132.6	1.2
PEA_30_2	13.2	0.6	30.7	9.3	168.2	6.9
PEA_35_2	15.5	0.8	26.7	7.8	199.2	3.0
PEA_37_2	14.1	4.4	13.0	5.9	221.0	9.9
PEA_41_2	11.6	2.5	6.4	1.1	240.2	4.0
PEA_44_2	10.7	1.9	4.7	1.0	268.8	8.3
PEA_48_2	9.5	3.5	3.7	1.2	295.3	17.0
PEA_50_2	6.3	1.3	3.1	0.8	251.5	38.4

Note, that the shown values are the average values of 5 measured samples, except PEA_18_2 and PEA_50_2, which consisted of only 4 and 2 samples, respectively, due to imperfection of the specimens.

35 %). The same behavior was found in the work of Hao *et al.* (2013), who found an optimum tensile strength for their aliphatic polyester amides at an amide content of 50 %. However, other sources revealed that an increase in amide content resulted in enhanced tensile strength [27,29,42].

The elongation behavior of the polyester amides is worth mentioning since one polymer (PEA_18_2) considerably outperformed all the other samples based on its ductility. The higher value for the strain at ultimate stress (233 ± 32 %) of this sample compared to the other polymers (ranging from 3 to 31 %) could be correlated with its relatively low T_g and limited degree of crystallinity. An inverse relationship is present between the T_g and the degree of crystallinity of the polyester amides and the ductility; a low T_g in combination with a relatively low degree of crystallinity resulted in a high elongation. These properties are attributed to the flexibility of the material, related to chain mobility.

The E-moduli, also referred to as the Young's moduli, showed a similar trend, in line with the stress and strain results. PBA demonstrated a high E-modulus, which subsequently reached a minimum for the polyester amide (PEA_18_2), which had the lowest degree of crystallinity and highest strain.

Thereafter, the E-modulus improved with increasing amide content, which is in line with the findings for similar polyester amides and expectations, due to an increased degree of crystallinity and more hydrogen bonding[27–29,42]. A one-to-one comparison of the mechanical properties of similar polyester amides compared to those of polyester amides in this work cannot be performed due to differences in testing conditions, e.g. specimen sizes and pulling speed.

4.1. Hydrophilicity

In the previous paragraphs, it was revealed that the amide content of the polyester amides had a strong influence on the polymeric properties. Hence, water contact angle measurements were performed to determine the influence of the amide content on the hydrophilicity, which might be valuable to correlate with biodegradability[46]. All polyester amides and PBA from the high $\overline{M_n}$ series contain a hydrophilic surface (WCA < 90°), as depicted in Fig. 7[47]. In addition, similar to the previous results, the minimum value corresponded to PEA_18_2, and indicated that PEA_18_2 was the most hydrophilic polymeric material, illustrated as insert in Fig. 7. Generally, all polyester amides were hydrophilic, with contact angles ranging from 69 to 87°. The inconsistency of the magnitude of the error bars was attributed to minimal surface imperfections. In addition, small irregularities in film thickness and slightly tilted surfaces could lead to different forces inside the droplet and therefore to minor deviations in water contact angles.

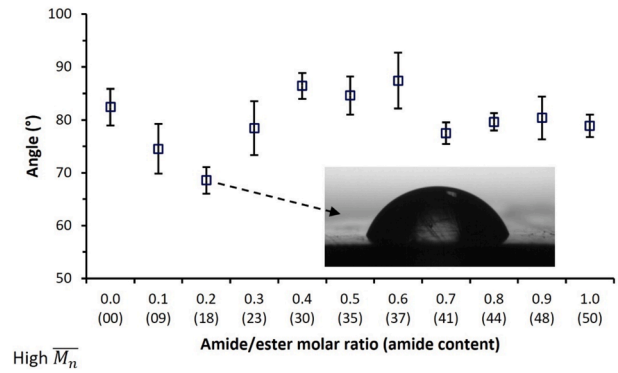

Fig. 7. Water contact angle versus the amide-to-ester ratio and amide content (%) of the high $\overline{M_n}$ polyester amides series, and picture of a water droplet (5 μL) at the surface of PEA_18_2.

4.2. Biodegradability

The biodegradability of polymers is a broadly defined concept, and there are numerous ways to demonstrate or estimate the biodegradability of polymers. Biodegradation using activated sludge is generally seen as a viable method to indicate the degradation of polymers by microorganisms due to its highly versatile ecosystem[48]. However, biodegradation is dependent not only on the presence and type of microorganism, but also on the molecular weight, degree of crystallinity, hydrophilicity, thermal transition points, polymer structure, particle size and environmental conditions[49,50].

The percentages of biodegradation, obtained from the BOD values, are depicted in Fig. 8a. The pure polyester (PBA) and the polyester amides with an amide content up to 23 % reached a relatively high degree of biodegradation, ranging from 41 to 69 %, within 28 days. The other polyester amides demonstrated a biodegradable character over

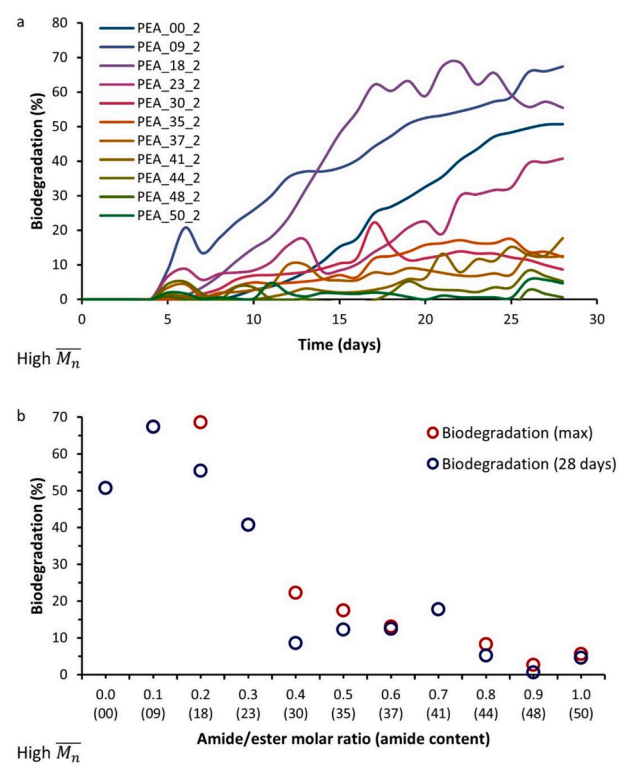

Fig. 8. (a) Percentage of biodegradation in activated sludge versus measurement time, and (b) versus the amide-to-ester ratio and amide content (%) of the high $\overline{M_n}$ polyester amide series.

time as well, however lower values were reached (3 to 22 %). These results are more promising than the patented data on aliphatic polyester amides, describing that degradation was realizable at high pH and elevated temperature (pH 12, 80 °C)[51].

The influence of the amide content on the biodegradability is visualized in Fig. 8b in more detail. The highest degree of biodegradation was obtained for the low amide content polyester amides. This can be attributed to the reduced T_g values, lower degree of crystallinity and lower water contact angles (higher hydrophilicity) of these polymers. In addition, Park *et al.* (2003) demonstrated that the addition of low amounts of amine groups (up to 20 %) to PBA improved biodegradability, which was attributed to the decreased regularity of the molecular structure and increased surface hydrophilicity[46]. Another noteworthy observation is the reduced biodegradability with increasing amide content. This is in accordance with expectations related to the higher degree of crystallinity and higher thermal transition points. Furthermore, enzymatic degradation of polyester amide showed that a higher amide content resulted in reduced weight loss over time[28].

These results confirm that aliphatic polyester amides not only are enzymatically degradable, hydrolysable under simulated physiological conditions, or degradable under harsh conditions, but show clear biodegradable behavior over time in microorganism-rich media as well [28,29,51]. In addition, the degree of biodegradation of these polyester amides can be tuned by simply varying the amide content.

5. Conclusion

Two series of polyester amides based on butanediol, adipic acid and AH salt were synthesized via a bulk polymerization process. The amide content was varied to study the influence on the polymeric properties.

This was clearly visualized by analyzing the thermal properties; the T_g increased linearly with increasing amide content and ranged from −51 to −19 °C for the whole series. A similar behavior was observed for the melting points of the polyester amides, which increased from 66 to 201 °C with increasing amide content. Generally, a higher amide content resulted in a relatively higher degree of crystallinity. The highest degree of crystallinity was observed for the pure polyester PBA, while the lowest was found for the polyester amide with an amide-to-ester ratio of 0.2: 1.0 (amide content of 18 %).

The amide content also significantly influenced the mechanical properties. The ultimate stress reached an optimum (15.5 MPa) for an amide content of 35 %, while that of the pure polyester reached a slightly higher value (15.9 MPa), which was assigned to the considerably higher degree of crystallinity and slightly higher molecular weight with respect to the polyester amides.

All polyester amides were shown to be hydrophilic, based on water contact angle measurements. Finally, the biodegradability tests demonstrated that the pure polyester PBA and all polyester amides were biodegradable over time in activated sludge. An improved biodegradation behavior was observed for the low amide content polyester amides with respect to the aliphatic polyester PBA. In addition, the biodegradability of these polyesters can be steered by altering the amide-to-ester ratio of these aliphatic polyester amides.

The results of this study confirm the high potential of these biobased and biodegradable polymers with tunable properties over a wide range, by the change in amide content. Therefore, these biobased polyester amides are highly interesting for multiple sustainable polymer applications, for example as packing materials, in agriculture and possibly in other engineering applications, due to their ease of production, strength, availability of starting materials and biodegradability.

CRediT authorship contribution statement

Cornelis Post: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jeroen van der Vlist:** Writing – review & editing, Methodology, Conceptualization. **Jesse A. Jongstra:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Rudy Folkersma:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Vincent S.D. Voet:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Katja Loos:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by Greenwise Campus, Regio Deals, Province of Drenthe, and Municipality of Emmen. We would like to acknowledge Patrizio Raffa and Cor Koning, from the Engineering and Technology Institute Groningen (ENTEG), University of Groningen, The Netherlands, to allow us to use their biodegradability test facilities. In addition, we want to express our gratitude to Katrien Bernaerts, from Maastricht University, The Netherlands, for kindly performing the GPC analysis in HFIP.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113594.

Data availability

Data will be made available on request.

References

[1] H. Nakajima, P. Dijkstra, K. Loos, The recent developments in biobased polymers toward general and engineering applications: polymers that are upgraded from biodegradable polymers, analogous to petroleum-derived polymers, and newly developed, Polymers (Basel) 9 (10) (2017) 1–26, https://doi.org/10.3390/polym9100523.
[2] R.M. Cywar, N.A. Rorrer, C.B. Hoyt, G.T. Beckham, E.Y.X. Chen, Bio-based polymers with performance-advantaged properties, Nat. Rev. Mater. 7 (2) (2022) 83–103, https://doi.org/10.1038/s41578-021-00363-3.
[3] R.P. Babu, K. O'Connor, R. Seeram, Current progress on bio-based polymers and their future trends, Prog. Biomater. 2 (1) (2013) 8, https://doi.org/10.1186/2194-0517-2-8.
[4] I.D. Posen, P. Jaramillo, W.M. Griffin, Uncertainty in the life cycle greenhouse gas emissions from U.S. production of three biobased polymer families, Environ. Sci. Technol. 50 (6) (2016) 2846–2858, https://doi.org/10.1021/acs.est.5b05589.
[5] A. Muthukumar, S. Veerappapillai, Biodegradation of plastics – A brief review, Int. J. Pharm. Sci. Rev. Res. 31 (2) (2015) 204–209.
[6] V. Goel, P. Luthra, G.S. Kapur, S.S.V. Ramakumar, Biodegradable/Bio-plastics: myths and Realities, J. Polym. Environ. 29 (10) (2021) 3079–3104, https://doi.org/10.1007/s10924-021-02099-1.
[7] Plastics Europe, "Plastics - the fast facts 2023," *2023*. https://plasticseurope.org/knowledge-hub/plastics-the-fast-facts-2023/.
[8] G.G.N. Thushari, J.D.M. Senevirathna, Plastic pollution in the marine environment, Heliyon 6 (8) (2020) e04709, https://doi.org/10.1016/j.heliyon.2020.e04709.
[9] Y. Chae, Y.J. An, Current research trends on plastic pollution and ecological impacts on the soil ecosystem: a review, Environ. Pollut. 240 (2018) 387–395, https://doi.org/10.1016/j.envpol.2018.05.008.
[10] V.M. Azevedo-Santos, et al., Plastic pollution: a focus on freshwater biodiversity, Ambio 50 (7) (2021) 1313–1324, https://doi.org/10.1007/s13280-020-01496-5.
[11] K. Schuhen, Defu He, Yongming Luo (Eds.): Microplastics in terrestrial environments: emerging contaminants and major challenges, Anal Bioanal Chem 413 (2021) 6199–6200, https://doi.org/10.1007/s00216-021-03598-9.
[12] K. Bucci, M. Tulio, C.M. Rochman, What is known and unknown about the effects of plastic pollution: a meta-analysis and systematic review, Ecol. Appl. 30 (2) (2020) 1–16, https://doi.org/10.1002/eap.2044.
[13] M. Padervand, E. Lichtfouse, D. Robert, C. Wang, Removal of microplastics from the environment. a review, Environ. Chem. Lett. 18 (3) (2020) 807–828, https://doi.org/10.1007/s10311-020-00983-1.

[14] D.N. Bikiaris, Nanocomposites of aliphatic polyesters: an overview of the effect of different nanofillers on enzymatic hydrolysis and biodegradation of polyesters, Polym. Degrad. Stab. 98 (9) (2013) 1908–1928, https://doi.org/10.1016/j.polymdegradstab.2013.05.016.
[15] S. Dhamaniya, H.S. Jaggi, M. Nimiya, S. Sharma, B.K. Satapathy, J. Jacob, Synthesis, characterization and biodegradation studies of chain-coupled polyesters based on tartaric acid, Polym. Int. 63 (4) (2014) 680–688, https://doi.org/10.1002/pi.4569.
[16] M. Barletta, et al., Poly(butylene succinate) (PBS): materials, processing, and industrial applications, Prog. Polym. Sci. 132 (2022) 101579, https://doi.org/10.1016/j.progpolymsci.2022.101579.
[17] S.A. Rafiqah, et al., A review on properties and application of bio-based poly (Butylene succinate), Polymers (Basel) 13 (9) (2021) 1–28, https://doi.org/10.3390/polym13091436.
[18] M.A. Woodruff, D.W. Hutmacher, The return of a forgotten polymer - Polycaprolactone in the 21st century, Prog. Polym. Sci. 35 (10) (2010) 1217–1256, https://doi.org/10.1016/j.progpolymsci.2010.04.002.
[19] N. Jacquel, et al., Synthesis and properties of poly(butylene succinate): efficiency of different transesterification catalysts, J. Polym. Sci. Part A Polym. Chem. 49 (24) (2011) 5301–5312, https://doi.org/10.1002/pola.25009.
[20] A. Kausar, Trends in graphene reinforced polyamide nanocomposite for functional application: a review, Polym. Technol. Mater. 58 (9) (2019) 917–933, https://doi.org/10.1080/25740881.2018.1563115.
[21] M. Shakiba, et al., Nylon—A material introduction and overview for biomedical applications, Polym. Adv. Technol. 32 (9) (2021) 3368–3383, https://doi.org/10.1002/pat.5372.
[22] N. Yamano, A. Nakayama, N. Kawasaki, N. Yamamoto, S. Aiba, Mechanism and characterization of polyamide 4 degradation by Pseudomonas sp, J. Polym. Environ. 16 (2) (2008) 141–146, https://doi.org/10.1007/s10924-008-0090-y.
[23] A. Rodriguez-Galan, L. Franco, J. Puiggali, Degradable poly(ester amide)s for biomedical applications, Polymers (Basel) 3 (1) (2011) 65–99, https://doi.org/10.3390/polym3010065.
[24] M. Winnacker, B. Rieger, Poly(ester amide)s: recent insights into synthesis, stability and biomedical applications, Polym. Chem. 7 (46) (2016) 7039–7046, https://doi.org/10.1039/c6py01783e.
[25] A. Rodríguez-Galán, L. Franco, and J. Puiggalí, Biodegradable poly (Ester Amide) S: Synthesis and applications. 2011.
[26] P. Ranganathan, C.W. Chen, S.P. Rwei, Y.H. Lee, S.K. Ramaraj, Methods of synthesis, characterization and biomedical applications of biodegradable poly (ester amide)s- A review, Polym. Degrad. Stab. 181 (2020) 109323, https://doi.org/10.1016/j.polymdegradstab.2020.109323.
[27] S. Liu, C. Li, J. Zhao, Z. Zhang, W. Yang, Synthesis and characterization of polyesteramides having short Nylon-6 segments in the main chains through polycondensation and chain extension, Polymer (Guildf) 52 (26) (2011) 6046–6055, https://doi.org/10.1016/j.polymer.2011.10.048.
[28] Y. Hao, M. Chen, J. Zhao, Z. Zhang, W. Yang, Synthesis and properties of polyesteramides having short nylon-610 segments in the main chains through polycondensation and chain extension, Ind. Eng. Chem. Res. 52 (19) (2013) 6410–6421, https://doi.org/10.1021/ie302879t.
[29] E. Armelin, L. Franco, A. Rodríguez-Galán, J. Puiggalí, Study on the degradability of poly(ester amide)s related to nylons and polyesters 6,10 or 12,10, Macromol. Chem. Phys. 203 (1) (2002) 48–58, https://doi.org/10.1002/1521-3935(20020101)203:1<48::AID-MACP48>3.0.CO;2-E.
[30] E. Grigat, R. Koch, R. Timmermann, BAK 1095 and BAK 2195: Completely biodegradable synthetic thermoplastics, Polym. Degrad. Stab. 59 (1–3) (1998) 223–226, https://doi.org/10.1016/s0141-3910(97)00174-2.
[31] R. Timmermann, R. Dujardin, and R. Koch, "Thermoplastically processible and biodegradable aliphatic polyesteramides," US005644020A, 1997.
[32] DSM Engineering Plastics, "Achieving higher bio-based content in DSM Arnitel® copolyesters: Results from testing 1,4-butanediol (BDO) made with Genomatica's process technology.," 2013, [Online]. Available: https://www.dsm.com/content/dam/dsm/arnitel/en_US/documents/DSM-Paper-Bio-BDO.pdf.
[33] A. Scott, Covestro partners to make bio-HDM, C&EN Glob. Enterp. 100 (3) (2022) 13, https://doi.org/10.1021/cen-10003-buscon12.
[34] Toray Innovation by Chemistry, "Toray Invents 100% Bio-Based Adipic Acid from Sugars Derived from Inedible Biomass, Scaling Up for Application to Eco-Friendly Nylon 66," 2023. https://www.toray.com/global/news/details/20220818103248.html (accessed Aug. 31, 2023).
[35] J.H. Chen, W.Q. Yuan, Y.D. Li, Y.X. Weng, J.B. Zeng, Malleable and Sustainable Poly(ester amide) networks synthesized via melt condensation polymerization, ACS Sustain. Chem. Eng. 7 (18) (2019) 15147–15153, https://doi.org/10.1021/acssuschemeng.9b03956.
[36] P.E. Beck, E.E. Magat, Poly(hexamethylene adipamide) by Melt Polymerization [Nylon 6,6], Macromol. Synth. 1 (1976) 248–250.
[37] OECD Guideline for tetsing of chemicals, "OECD 301 - Ready Biodegradability," *OECD Guidel. Test. Chem.*, 1992.
[38] C. Post, et al., Enzymatic bulk synthesis, characterization, rheology, and biodegradability of biobased 2,5-bis(hydroxymethyl)furan polyesters, Green Chem. 26 (15) (2024) 8744–8757, https://doi.org/10.1039/d4gc01512f.
[39] L. Papadopoulos, et al., Unlocking the potential of furan-based poly(ester amide)s: an investigation of crystallization, molecular dynamics and degradation kinetics of novel poly(ester amide)s based on renewable poly(propylene furanoate), Polym. Chem. 12 (38) (2021) 5518–5534, https://doi.org/10.1039/d1py00713k.
[40] K. Hemmrich, et al., Three-dimensional nonwoven scaffolds from a novel biodegradable poly(ester amide) for tissue engineering applications, J. Mater. Sci. Mater. Med. 19 (1) (2008) 257–267, https://doi.org/10.1007/s10856-006-0048-3.
[41] T.H.N. Nguyen, F. Balligand, A. Bormann, V. Bennevault, P. Guégan, Synthesis of new biobased linear poly(ester amide)s, Eur. Polym. J. 121 (2019) 109314, https://doi.org/10.1016/j.eurpolymj.2019.109314.
[42] P.A.M. Lips, R. Broos, M.J.M. Van Heeringen, P.J. Dijkstra, J. Feijen, Synthesis and characterization of poly (ester amide) s containing crystallizable amide segments, Polymer 46 (19) (2005) 7823–7833, https://doi.org/10.1016/j.polymer.2005.07.013.
[43] S. Bose, M. Mukherjee, T. Rath, C.K. Das, A.K. Saxena, Effect of polyphosphazene on the properties of blend of nylon 6,6/Thermotropic liquid crystalline polymer, J. Reinf. Plast. Compos. 28 (2) (2009) 157–166, https://doi.org/10.1177/0731684407084121.
[44] L. Hua, X. Wang, Polymorphic Crystallization Behavior of a Poly(butylene adipate) Midblock within a Poly(L-lactide-butylene adipate-L-lactide) Triblock Copolymer, Polymers (Basel) 14 (22) (2022) 4902, https://doi.org/10.3390/polym14224902.
[45] W. Zhang, J. Li, Y. Shang, H. Li, S. Jiang, L. An, Temperature dependence of tensile behavior in poly(butylene terephthalate) with different crystallinity, Mater. Des. 129 (April) (2017) 143–150, https://doi.org/10.1016/j.matdes.2017.04.035.
[46] C. Park, E.Y. Kim, Y.T. Yoo, S.S. Im, Effect of hydrophilicity on the biodegradability of polyesteramides, J. Appl. Polym. Sci. 90 (10) (2003) 2708–2714, https://doi.org/10.1002/app.12925.
[47] K.-Y. Law, Definitions for Hydrophilicity, Hydrophobicity, and Superhydrophobicity: Getting the Basics Right, J. Phys. Chem. Lett. 5 (4) (2014) 686–688, https://doi.org/10.1021/jz402762h.
[48] P. Torena, M. Alvarez-Cuenca, M. Reza, Biodegradation of polyethylene terephthalate microplastics by bacterial communities from activated sludge, Can. J. Chem. Eng. 99 (S1) (2021) S69–S82, https://doi.org/10.1002/cjce.24015.
[49] C. Post, D. Maniar, V.S.D. Voet, R. Folkersma, K. Loos, Biobased 2,5-Bis(hydroxymethyl)furan as a Versatile building block for sustainable polymeric materials, ACS Omega 8 (10) (2023) 8991–9003, https://doi.org/10.1021/acsomega.2c07629.
[50] Y. Tokiwa, B.P. Calabia, C.U. Ugwu, S. Aiba, Biodegradability of plastics, Int. J. Mol. Sci. 10 (9) (2009) 3722–3742, https://doi.org/10.3390/ijms10093722.
[51] J.-S. Lai, "Biodegradable polyesteramide and preparation method," US007196127B2, 2007.